Nilaa Raghunathan

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EDUCATION

Columbia University — **New York, US**
MS in Data Science — Sep 2024 - Dec 2025
- Coursework: Generative AI, Deep Learning, Machine Learning, Data Structures & Algorithms, Probability & Statistics.

Vellore Institute of Technology — **Vellore, IN**
BTech in Computer Science with Specialization in Data Science — Sep 2020 - May 2024
- Coursework: NLP, Artificial Intelligence, Business Intelligence, Cloud Computing, Data Analytics, Software Engineering.

TECHNICAL SKILLS

Languages: Python, SQL, Scala, C, C#, Java, TypeScript.
Distributed Systems: Apache Spark, PySpark, Kafka, Airflow, Databricks, Synapse, PostgreSQL, MySQL, Redis, Delta Lake.
Cloud & DevOps: Azure, AWS, Docker, Kubernetes, Git, Github Copilot, CI/CD, Jenkins, Linux, Azure Data Explorer (Kusto).
AI/ML: PyTorch, TensorFlow, scikit-learn, LLMs, RAG, Hugging Face, LangChain, FastAPI, PEFT, MLflow, Azure AI Foundry

EXPERIENCE

Limex Trading — **New York, US**
Applied AI Intern — **Sep 2025 - Dec 2025**
- Built an agentic financial research copilot combining Hybrid RAG, GraphRAG, LangGraph orchestration, and multi-hop knowledge retrieval, reducing analyst research time by 42%.
- Improved retrieval precision by 24% & Recall@10 by 18% using vector search, BM25, graph traversal & cross-encoder reranking.
- Engineered a financial knowledge graph with 35K+ nodes & 120K+ relationships, using schema-constrained extraction and confidence-scored validation for data quality and traceability.
- Fine-tuned Llama 3.1-8B with QLoRA on 600+ financial instruction pairs, improving citation accuracy from 72% to 81%.

Magna International Inc. — **Michigan, US**
Data Science Intern — **Jun 2025 - Aug 2025**
- Built an ML-based inventory forecasting pipeline using Azure Synapse and Databricks across 1,700+ automotive parts, reducing safety stock by 12% and saving $1.4M annually.
- Increased forecast accuracy by 18% (MAPE) using probabilistic time-series modeling (GMM-HMM) to capture demand patterns.
- Strengthened demand signal reliability by building a context-aware NLP pipeline (BiLSTM-RoBERTa) over 7K+ supplier records.
- Lowered stock-out risk by 13% through ANOVA, A/B testing & causal impact analysis, validating changes before deployment.
- Deployed a Streamlit application backed by Databricks and MySQL, enabling planners to monitor forecasts and inventory risk.

Vellore Institute of Technology — **Vellore, IN**
Applied Research Intern — **Apr 2022 - Jul 2023**
- Architected a distributed data platform using Kafka and Spark for scalable ingestion and transformation of 50M+ records.
- Built automated Airflow ETL pipelines with schema validation, data quality checks and monitoring to improve pipeline reliability.
- Published findings as first-author IEEE Access research on large-scale sentiment analysis across multilingual, cross-domain, and low-resource datasets, receiving 85+ citations.

PROJECTS

Multi-Agent HR Intelligence Platform ***(github.com/Nilaa-Jaya/Multi-Agent-HR-Intelligence-Platform-)*** — **May 2026 - Present**
- Architected and deployed multi-agent HR support system using LangGraph/LangChain with 9 specialized agents, implementing 7-way conditional routing and stateful orchestration to automate 85% of HR inquiries with 90%+ resolution accuracy.
- Built scalable RAG infrastructure with FAISS vector search and re-ranking, reducing query handling time by 30% and errors by 40% while maintaining <1s retrieval latency.
- Developed production FastAPI microservices with PostgreSQL analytics, Docker deployment, GitHub Actions CI/CD (38 tests), and secure webhooks for reliable system integration.

Generative Recommender System ***(github.com/Nilaa-Jaya/Generative-Recommender-System)*** — **Feb 2026 - May 2026**
- Designed RAG-based personalized-recommender system processing 230M+ reviews, 54M+ users & 1.57M+ products.
- Built scalable vector search system using Sentence-BERT, FAISS, diversity re-ranking, and persona-based retrieval via DeepMF embeddings, improving Precision@10 from 0.57 to 0.69 and Diversity@10 from 0.42 to 0.51 with ~43ms latency.
- Fine-tuned LLaMA2-7B using QLoRA with RLHF reward model on 100K pairs, improving BERTScore from 0.79 to 0.86

Financial Fraud Detection ***(github.com/Nilaa-Jaya/Financial-Fraud-Detection)*** — **Aug 2025 - Oct 2025**
- Developed fraud detection pipeline on 6.3M+ transactions, engineering error-based and anomaly features using Isolation Forest.
- Boosted fraud detection by ADASYN oversampling with ML (Logistic Regression, Random Forest, XGBoost) & deep learning models (RNN, LSTM, BiLSTM) achieving 0.98 F1-score, 97% recall & reducing false negatives by 50%.

PUBLICATIONS

- Challenges and Issues in Sentiment Analysis: Comprehensive Survey | *IEEE Access (Cited by 85+)*.
- Improved Deep Learning Model for Sentiment Analysis using LSTM with Layer Normalization | *IEEE - AAIAC Conference*.